

Healthcare Appointment Reminder API

Production-Grade Product, Integration & Compliance Design

PREPARED FOR THE VAN-KAAL × ZOCDOC PILOT

Purpose: Convert the no-show reminder pilot into a reusable API platform that customer applications can integrate once and then use for appointment reminders, patient confirmation/cancellation, IVR calls, status webhooks and usage-based billing.

External milestone: 28 September 2026 — The supplied pilot specification separately lists 26 September for the live demo and 16–18 September for architecture finalization.

Document status: Proposed production architecture. Client decisions are clearly marked as OPEN. This document is an engineering/product design and compliance-readiness guide; legal counsel should confirm contractual and regulatory obligations.

1. Executive Summary

The recommended product model is simple: the customer application sends appointment details to Van-Kaal once when an appointment is created. Van-Kaal becomes responsible for scheduling the communication workflow, sending the patient reminders, collecting confirmation/cancellation, and notifying the customer system of the resulting status.

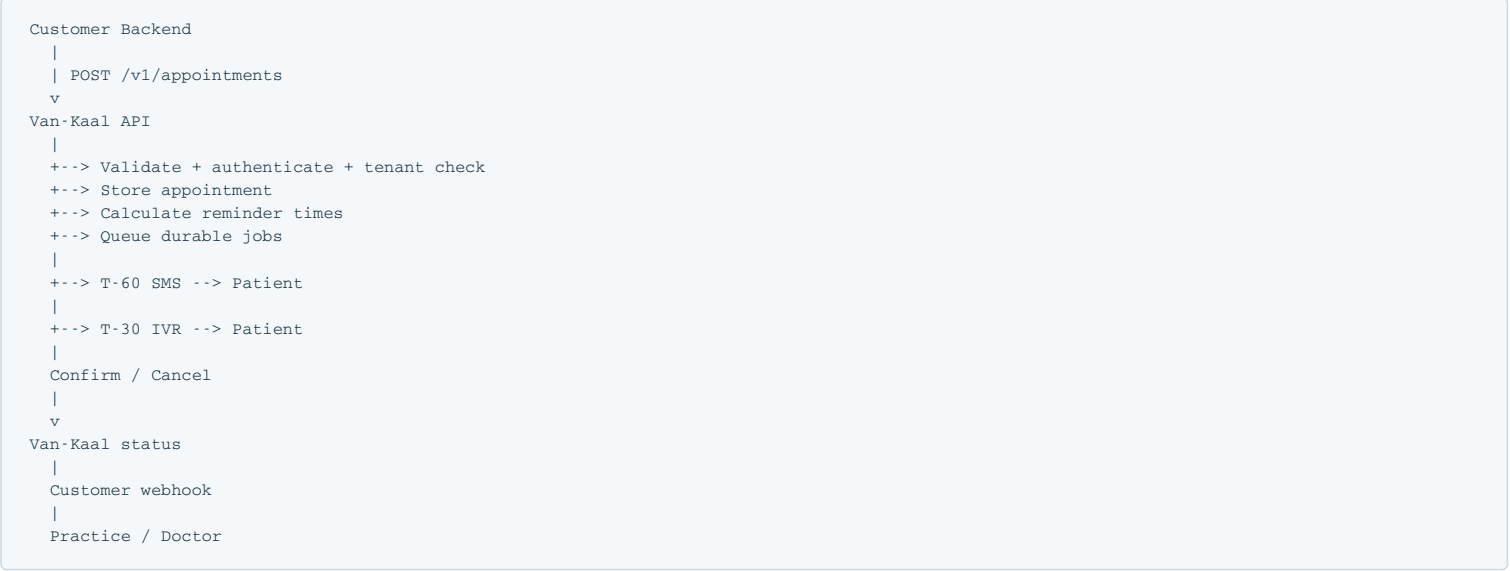
CORE DESIGN DECISION
The customer application should call the Van-Kaal API when the appointment is created — not 30 minutes before the appointment. Van-Kaal owns the reminder schedule. This is what makes the platform reusable and allows Van-Kaal to handle late bookings consistently.

Layer	Responsibility
Customer system	Create/update/cancel/reschedule appointments; expose a secure webhook endpoint; surface patient status to staff/doctor.
Van-Kaal API	Validate requests, authenticate client, apply tenant rules, store necessary data, schedule reminders, expose appointment status.
Van-Kaal Communication Engine	Decide when to send SMS and IVR, handle late bookings, retry failures, prevent duplicate sends.
Patient channels	Receive SMS and/or voice reminder; confirm or cancel by reply, button/link, or IVR input.
Customer webhook	Receive trusted appointment outcome events such as confirmed, cancelled, at-risk, delivery failure and call completion.
Billing/usage	Meter billable SMS and call usage without unnecessarily copying PHI into billing records.

2. Product Concept — How the Service Works

The current pilot specification establishes the underlying workflow: ZocDoc provides appointment data, Van-Kaal stores appointment/reminder/response data, sends SMS and IVR reminders, and exposes practice-facing status and analytics. The target timing in the current pilot document is 12:00 PM SMS and 1:30 PM IVR for a 3:30 PM appointment. This document proposes the 60-minute SMS / 30-minute IVR model described by the client; that timing is an OPEN client decision until approved.

PROPOSED STANDARD REMINDER POLICY
SMS at T-60 minutes; IVR at T-30 minutes; confirmations/cancellations immediately update the appointment state and generate an outbound webhook to the customer system.



3. What Changes Are Required in the Customer Codebase?

IMPORTANT SECURITY RULE

The Van-Kaal API credential must be used from the customer backend/server, not directly from a public web or mobile frontend. The frontend should call the customer backend; the customer backend calls Van-Kaal.

Customer-side change	What they implement	Why it is needed
1. Appointment create integration	After a booking is successfully created, call POST /v1/appointments with appointment ID, patient contact, date/time, timezone, doctor and practice information.	Starts the Van-Kaal reminder workflow.
2. Stable appointment ID	Send a unique ID from the source system.	Enables idempotency, retries and reconciliation.
3. Timezone	Send an IANA timezone such as America/New_York.	Prevents reminder times being wrong across US time zones and daylight-saving changes.
4. Cancel integration	When an appointment is cancelled, call the cancel endpoint.	Stops future reminders and keeps Van-Kaal status synchronized.
5. Reschedule integration	When time/date changes, call the reschedule/update endpoint.	Recalculates future reminder jobs instead of sending stale reminders.
6. Customer webhook endpoint	Expose a server endpoint such as /webhooks/vankaal that accepts signed Van-Kaal events.	Lets Van-Kaal push patient outcomes back to the customer system.
7. Webhook verification	Verify Van-Kaal signature, timestamp and event ID before processing.	Prevents spoofed status updates.
8. Staff/doctor display	Update the practice/doctor dashboard using webhook events or the customer system's own data store.	Keeps the doctor informed without coupling Van-Kaal to the customer UI.
9. Credential storage	Store Van-Kaal credentials in the customer server secret store.	Prevents credentials from appearing in source code or browser bundles.
10. Error/retry handling	Treat 2xx as accepted; retry safely on transient 5xx/network failures using the idempotency key.	Avoids duplicate appointments and missed reminders.

3.1 Minimal Customer Integration

```
// Customer BACKEND example
POST https://api.vankaal.com/v1/appointments
Authorization: Bearer <server-side-api-key>
Idempotency-Key: booking-APT-98231-v1
Content-Type: application/json

{
  "appointment_id": "APT-98231",
  "patient": {
    "name": "John Doe",
    "phone": "+14155551234"
  },
  "appointment": {
    "date": "2026-09-28",
    "time": "15:30:00",
    "timezone": "America/New_York",
    "doctor": "Dr. Michael Smith",
    "practice_id": "practice_001"
  }
}
```

3.2 API Response to the Customer

```
{
  "success": true,
  "appointment_id": "APT-98231",
  "status": "scheduled",
  "reminders": {
    "sms": { "status": "scheduled", "scheduled_for": "2026-09-28T14:30:00-04:00" },
    "voice": { "status": "scheduled", "scheduled_for": "2026-09-28T15:00:00-04:00" }
  }
}
```

This response is an acknowledgement that Van-Kaal accepted and scheduled the workflow. It is not the future patient confirmation. Patient outcomes arrive asynchronously through the configured customer webhook.

4. Recommended API Surface (v1)

Method	Endpoint	Purpose
POST	/v1/appointments	Create/register an appointment and schedule reminders.
GET	/v1/appointments/{appointment_id}	Read current Van-Kaal status and reminder state.
POST	/v1/appointments/{appointment_id}/cancel	Cancel appointment and suppress future reminders.
POST	/v1/appointments/{appointment_id}/reschedule	Replace appointment time/date/timezone and recalculate reminder jobs.
POST	/v1/appointments/{appointment_id}/reminders/retry	Optional controlled retry for eligible failed reminder operations.
GET	/v1/usage	Usage for billing/reconciliation.
POST	/v1/webhook-test	Optional endpoint for validating a customer webhook configuration.

API VERSIONING
Use /v1 from day one. Avoid breaking changes inside the same version. Add fields compatibly, deprecate clearly, and publish a versioned integration guide.

5. Patient Confirmation & Doctor/Practice Webhook

The cleanest architecture is asynchronous. The patient action changes the Van-Kaal appointment state; Van-Kaal then sends a signed webhook to the customer system. The customer system decides how to display the result to the doctor or practice staff.

Patient outcome	Van-Kaal event	What customer system should do
Confirmed by SMS reply	appointment.confirmed	Mark appointment confirmed; show confirmation timestamp to staff/doctor.
Confirmed by secure link/button	appointment.confirmed	Same business result; store source as link/button if useful.
Confirmed by IVR	appointment.confirmed	Same business result; store channel = voice.
Cancelled by SMS/link/IVR	appointment.cancelled	Mark appointment cancelled; apply client policy for slot release.
No response after configured workflow	appointment.at_risk	Flag for staff follow-up if the client wants this behavior.
SMS delivery failure	reminder.sms_failed	Surface delivery issue for operational visibility; do not assume patient received it.
IVR failed/not answered	reminder.voice_failed	Record outcome; apply retry policy if enabled.
Reminder successfully sent	reminder.sent	Update communication history and usage meter.

5.1 Example Webhook

```
POST /webhooks/vankaal
X-VanKaal-Signature: <HMAC-SHA256>
X-VanKaal-Timestamp: 178...
X-VanKaal-Event-Id: evt_01J...
Content-Type: application/json

{
  "event": "appointment.confirmed",
  "event_id": "evt_01J...",
  "appointment_id": "APT-98231",
  "status": "confirmed",
  "channel": "sms_reply",
  "occurred_at": "2026-09-28T14:37:12-04:00"
}
```

WEBHOOK SECURITY
The customer must verify the signature and reject stale/replayed requests. Van-Kaal should retry non-2xx responses with backoff and include a unique event_id so the customer can process each event idempotently.

6. Reminder Decision Engine & Late Booking Rules

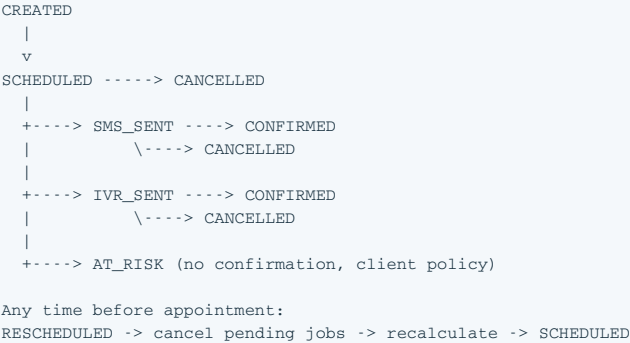
Late bookings are unavoidable in real scheduling systems. Van-Kaal should never require the customer application to solve them. The API accepts the booking and the Van-Kaal scheduler chooses the correct action based on the remaining time.

Booking timing	SMS behavior	IVR behavior	Recommended state
More than 60 min before appointment	Schedule for T-60	Schedule for T-30	scheduled
30-60 min before appointment	Send immediately	Schedule for T-30 / or immediate if policy says so	late_booking
5-30 min before appointment	Send immediately	Send immediately, subject to minimum voice lead time	late_booking
Less than minimum voice lead time	Send immediately if enabled	Skip or send only if client policy allows	late_booking / voice_skipped
Appointment already started/past	Do not send	Do not send	too_late
Appointment already cancelled	Do not send	Do not send	cancelled

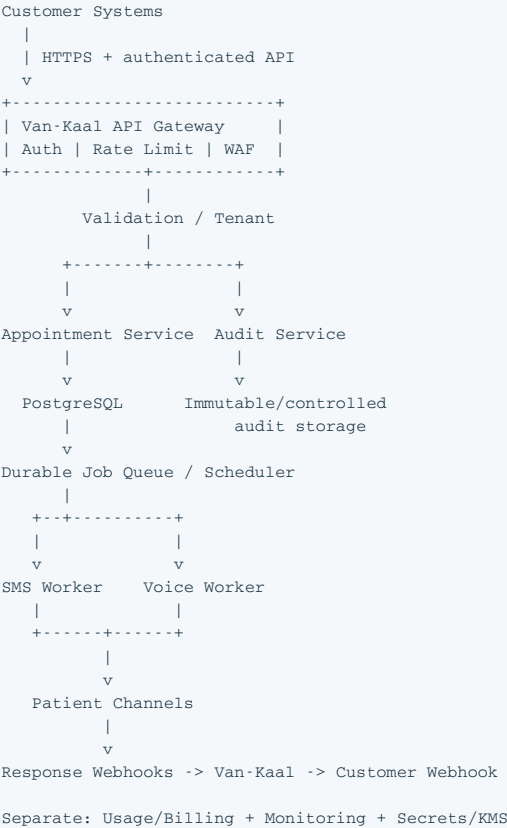
RECOMMENDED DEFAULT

Use a configurable minimum voice lead time (for example, 5 minutes) rather than calling a patient for an appointment that is only a minute away. The actual threshold must be confirmed with the client.

6.1 Booking/Appointment State Machine



7. Production-Grade Architecture



7.1 Why a Durable Queue Is Required

Do not depend on in-memory timers, application-server cron processes, or a single Node.js instance to guarantee a reminder. Production scheduling requires durable jobs that survive process restarts, deployments and partial failures. Workers should claim jobs safely, record delivery attempts, and prevent duplicate sends.

Production concern	Required behavior
Retries	Retry transient provider/network errors with exponential backoff and a maximum attempt count.
Idempotency	The same appointment create/cancel/reschedule request must be safe to retry without duplicating reminders.
Job locking	A reminder job must have a single owner at execution time, with lease/visibility timeout semantics.
Provider callbacks	Verify provider signatures and correlate callbacks to the correct reminder/job.
Dead-letter handling	Failed jobs that exceed retry limits go to a reviewable dead-letter queue.
Clock/timezone	Use UTC internally; retain the source IANA timezone for schedule calculations and display.
Reschedule	Remove/supersede pending jobs when appointment time changes.
Cancellation race	Before sending, re-check appointment status so a just-cancelled appointment is not contacted.
Deployment safety	Graceful worker shutdown and job handoff during releases.
Reconciliation	Periodic reconciliation can identify appointments whose scheduled reminder events do not match expected state.

8. Security & HIPAA Readiness

The supplied pilot specification already calls for encryption at rest, TLS 1.2+, limited PHI logging, PostgreSQL storage and practice-only access. For production, these controls need concrete implementation, ownership and evidence. [HHS-1]

Control	Production requirement	Evidence to retain
BAA / contracting	Confirm Van-Kaal role and contracting path; execute required BAA(s) before PHI processing where applicable. Downstream vendors that handle ePHI must also be contractually addressed. [HHS-1, HHS-3]	Executed agreements; subprocessor list; responsibility matrix
Risk analysis	Document threats, vulnerabilities, safeguards, owners and residual risk. HHS describes risk analysis as foundational. [HHS-2]	Risk assessment; remediation tracker; review history
Encryption	Encrypt ePHI in transit and at rest; define algorithms/settings with cloud/provider choices.	Architecture/configuration evidence; key policy
Key management	Use managed KMS/secrets storage; restrict access; rotate keys/credentials; log administrative access.	Key policy; access logs; rotation records
Access control	Unique user IDs, least privilege, RBAC, practice/tenant isolation, MFA for staff/admin access.	RBAC matrix; MFA configuration; access review records
Audit controls	Log security-relevant access and changes; protect logs from unauthorized alteration/deletion.	Audit schema; log samples; retention controls; review evidence
PHI minimization	Only transmit/store fields necessary for the reminder function. HHS minimum-necessary guidance supports limiting unnecessary PHI access/use. [HHS-6]	PHI inventory and field justification
Backups / recovery	Encrypted backups; documented restore testing; defined RPO/RTO. HHS notes backup/data recovery as relevant to cloud/security risk management. [HHS-4]	Backup policy; restore test results
Incident response	Formal detection, triage, containment, investigation, breach assessment, notification and post-incident steps. HHS generally sets a 60-calendar-day outer limit for applicable business-associate notification, while contracts may require faster internal escalation. [HHS-5]	IR plan; incident templates; escalation contacts
Vendor management	Confirm cloud, SMS and voice providers' healthcare/security terms and applicable BAA coverage.	Subprocessor register; contracts/BAAAs; risk reviews

IMPORTANT COMPLIANCE WORDING

Avoid a blanket "HIPAA certified" claim. HHS states that the Security Rule does not require certification and that private certifications do not remove the underlying legal obligations. [HHS-8]

8.1 SMS / Voice Privacy Design

HHS states that appointment reminders are permitted communications under the HIPAA Privacy Rule, but reasonable safeguards and appropriate limitation of information still matter. [HHS-7] The exact patient-facing content and communication vendors should therefore be client-approved before production.

- Prefer minimum necessary content in SMS and voice scripts; do not include diagnoses, medical history, insurance details or unrelated PHI.
- Confirm whether the required channel is SMS only, SMS + another supported channel, or a specific "iMessage" capability. The provider and healthcare contractual support must be confirmed.
- Use secure, expiring, non-guessable tokens for Confirm/Cancel links. Do not expose raw patient identifiers or appointment IDs in

- URLs.
- Treat provider delivery failures as operational events, not as proof that the patient did or did not see a message.
 - Define opt-out, wrong-number and shared-phone handling with the client and communications provider.

9. Data Model & Tenant Isolation

Entity	Key fields	Notes
Tenant	tenant_id, name, status	Top-level customer boundary.
Practice	practice_id, tenant_id, timezone, status	A customer may have multiple practices.
Appointment	appointment_id, practice_id, patient_ref, appointment_time, timezone, status	Business object; must be tenant-scoped.
Reminder	reminder_id, appointment_id, channel, scheduled_at, sent_at, status, provider_ref	Tracks SMS/voice workflow.
Patient contact	contact_ref, phone, optional name	Minimize stored fields; consider tokenized/internal references where possible.
Patient response	response_id, appointment_id, source, response, received_at	Record confirmation/cancellation outcome.
Audit event	event_id, tenant_id, actor/service, action, resource, timestamp, result	Designed for compliance and investigations.
Usage event	usage_id, tenant_id, event_type, quantity, timestamp	Keep billing usage separate from unnecessary PHI.
Webhook event	event_id, tenant_id, type, status, attempt_count, delivered_at	Supports retries and customer reconciliation.

TENANT ISOLATION

Every request and database query involving appointment or patient data must be tenant-scoped on the server. Frontend filtering alone is not a security boundary.

10. Usage-Based Billing Model

The API model supports the original business idea of charging customers according to usage. The cleanest commercial model is a platform fee plus metered communication usage, with optional enterprise features.

Billable unit	Metering event	Suggested metric
SMS	sms.sent or provider-confirmed billable message	Message segments/units, subject to provider billing model
Voice	voice.call.connected / duration	Seconds or billable minutes, depending on contract/provider
API	api.request.accepted	Optional; often better included in platform allowance
Premium features	feature.usage	Optional: analytics, advanced routing, extra channels, reporting
Platform	subscription.active	Monthly practice/customer fee

BILLING INTEGRITY

A billing event should have a unique usage_event_id and enough operational metadata to reconcile against provider invoices, but it should not duplicate unnecessary PHI into the billing system.

11. Reliability, Monitoring & Operations

Area	Production requirement
Availability	Target the pilot document's 99.5%+ uptime goal, then define the formal SLA separately. <i>[Pilot Spec, Section 6.2]</i>
Metrics	API latency, error rates, queue depth, jobs due/late, send success rate, provider failures, webhook success, confirmation rate.
Alerts	Repeated provider failures, queue backlog, auth failures, abnormal API activity, webhook failure spikes, database issues.
Tracing	Correlation ID from customer request through scheduler, reminder and webhook.
Operational dashboard	Per-tenant appointment and communication health without exposing unnecessary PHI.
Deployments	CI/CD, automated tests, database migrations, rollback plan, environment separation.
Backups	Scheduled encrypted backups plus periodic restore validation.
Support access	Time-bounded, approved, audited access; no informal production database browsing.
Disaster recovery	Document RPO/RTO, dependency recovery order and communications process.

12. Open Questions — Client Confirmation Required

These questions should be closed before the production architecture is considered locked. The recommended default column is an engineering proposal only; the client/business/legal owner must confirm the final decision.

Priority	Question	Why it matters	Recommended starting position
P0	BAA / relationship	Who is Van-Kaal contracting with for PHI processing: the practice directly, ZocDoc, or another entity? Who supplies/signs the BAA?	Do not process real PHI in production until the contractual path is confirmed.
P0	Reminder timing	Confirm standard policy: SMS T-60 and IVR T-30. The supplied pilot document currently specifies a different timing.	Approve 60/30 as the product default if that is the intended requirement.
P0	Late booking policy	What should happen when booking occurs <60, <30, <5 minutes before the appointment?	Recommended: SMS immediately; IVR immediately only when enough lead time remains.
P0	Messaging channel	Is "iMessage" a hard requirement, or is SMS sufficient for the pilot? Which approved provider will be used?	Choose a provider whose capabilities, contract and healthcare requirements are acceptable to the client.
P0	Voice provider	Which telephony provider will place IVR calls?	Provider must be approved and contractually/security reviewed before PHI use.
P0	Patient message content	Approve exact SMS and voice scripts. Can doctor and practice names be spoken/displayed?	Use minimum necessary content and client-approved language.
P0	Confirmation mechanism	Should patients be allowed to confirm by reply, link/button, IVR, or all three?	Support multiple channels if commercially and technically approved.
P0	Cancellation behavior	When patient cancels, should Van-Kaal only notify the customer, or is Van-Kaal expected to call the source booking API to cancel?	Recommended: customer system remains system of record; Van-Kaal notifies via webhook.
P0	Rescheduling	Can appointments be rescheduled, and how quickly must reminders be recalculated?	Recommended: customer calls reschedule endpoint; Van-Kaal supersedes old pending jobs.
P0	Webhook owner	What URL will receive Van-Kaal events? Which system is the system of record for appointment status?	Customer backend owns webhook endpoint and doctor-facing UI.
P0	Webhook response contract	Which events are required, and what retry window is acceptable?	Start with confirmed, cancelled, at_risk, reminder sent/failed, call outcome.
P0	Retention	What operational PHI retention period is required by contract/client/state law? What must remain for compliance documentation/audit purposes?	Do not use a single "90 days" rule for every data category.
P0	Data residency/access	Can production PHI be hosted in the selected region, and can Van-Kaal support engineers access production data?	Prefer least-privilege production access and document any cross-border access requirements.
P1	Practice hierarchy	Will one customer have multiple practices/locations?	Design multi-tenant from day one.
P1	Volume	Expected appointments/day, peak appointments/minute, practices, call minutes/day?	Needed for capacity and provider sizing.
P1	SLA	Confirm 99.5%+ uptime target and define maintenance/exclusions.	Separate pilot success metric from contractual SLA.
P1	Dashboard	Is Van-Kaal dashboard required for pilot, or is webhook integration sufficient?	Webhook should be authoritative integration path; dashboard can be optional operational UI.
P1	International patients	Will only US numbers/timezones be supported for the pilot?	Scope to US initially unless expansion is required.
P1	Opt-out/wrong number	What patient preference, opt-out and wrong-number handling is required?	Define with legal/compliance + communication provider.
P1	Testing data	Will client provide a sandbox/test tenant and synthetic patient records?	Use synthetic data for testing wherever possible.

13. Production Readiness & Evidence Pack

Deliverable	Owner	Status before Sep 28
Architecture diagram + data flow	Engineering	Required
API specification + examples	Engineering	Required
Webhook specification	Engineering	Required
PHI data inventory / field mapping	Engineering + Compliance	Required
BAA / contract responsibility map	Business + Legal	Client confirmation required
Subprocessor/vendor register	Engineering + Compliance	Required
Security risk analysis	Security/Compliance	Required for production readiness
Access control/RBAC matrix	Engineering/Security	Required
MFA configuration	Engineering/Security	Required for staff/admin access
Encryption/KMS/key policy	Engineering/Security	Required

Audit log specification + retention	Engineering/Security	Required
Incident response / breach workflow	Security/Legal	Required
Retention/deletion policy	Engineering + Compliance	Client confirmation required
Backup/restore evidence	Engineering	Required
Load/reliability test results	Engineering	Required
Penetration/security testing scope	Security	Recommended/confirm scope
Pilot runbook + rollback plan	Engineering/Operations	Required
Client integration guide	Engineering	Required
Usage/billing reconciliation report	Engineering/Finance	Required for commercial pilot

14. Recommended Delivery Plan to 28 September 2026

Period	Focus	Exit criteria
17-18 Sep	Architecture lock + client decisions	Open P0 questions resolved or explicitly accepted as blockers; API/webhook contract frozen.
19-21 Sep	Core API + database + tenant/auth + durable scheduling	Appointment create/cancel/reschedule works with idempotency and durable jobs.
22-23 Sep	SMS + IVR provider adapters + patient actions	Real provider flow works in controlled environment; retries and provider callbacks handled.
24 Sep	Webhooks + practice integration + audit + usage	Customer receives signed events; audit trail and metering operational.
25 Sep	Security hardening + failure scenarios	Late bookings, cancellations, reschedules, duplicate requests, provider failures and incident drills tested.
26 Sep	Live end-to-end demo rehearsal / demo	Matches pilot flow; current specification lists 3-5 sample appointments and zero crashes during the demo window. <i>[Pilot Spec, Section 6.1]</i>
27 Sep	Final production checklist	No P0 technical blockers; rollback and support plan ready.
28 Sep	ZocDoc presentation / pilot discussion	Demonstrate integration, patient workflow, operational controls and evidence pack.

15. Production-Grade Acceptance Criteria

Category	Acceptance criterion
API	Customer backend can create, cancel and reschedule appointments securely.
Scheduling	Reminder jobs survive restarts/deployments and execute once according to configured policy.
Late bookings	Defined and tested behavior exists for 60, 30, 15, 5 and 0-minute lead times.
Patient actions	SMS reply/link and IVR actions map to the correct appointment and update status safely.
Webhooks	Customer receives signed, idempotent outcome events with retry behavior.
Tenant isolation	A customer/practice cannot access another tenant's PHI through API, database or dashboard paths.
Security	Secrets are not stored in frontend bundles/source; access is authenticated and least-privilege.
Audit	Security-relevant activity is logged and protected against unauthorized alteration/deletion.
Compliance	BAA/vendor responsibilities, risk analysis and retention approach are documented before production PHI use.
Operations	Monitoring, alerting, backups, restore testing and incident-response procedures are in place.
Billing	Usage events can be reconciled to provider activity without requiring PHI in billing records.
Demo	End-to-end flow works from customer booking → reminder → patient response → customer webhook.

16. Final Architecture Decisions to Lock

- 1 Van-Kaal is an API-first platform; the customer application integrates through the customer backend.
- 2 The customer calls Van-Kaal when an appointment is created; Van-Kaal owns reminder scheduling.
- 3 Default reminder policy is proposed as SMS T-60 and IVR T-30, subject to client approval.
- 4 Late-booking logic is handled by Van-Kaal, with configurable thresholds and no scheduling of jobs in the past.
- 5 Customer system remains the appointment system of record; Van-Kaal sends outcome webhooks rather than silently changing the customer record unless explicitly integrated to do so.
- 6 Webhooks are signed, timestamped, retried, and idempotent.
- 7 Appointments, reminders, responses and audit events are tenant-scoped.
- 8 Production scheduling uses durable jobs/queues, not in-memory timers.

- 9
- Billing is usage-metered separately from PHI data and supports SMS/call reconciliation.
- 10
- Production claims about HIPAA readiness are backed by contracts, risk assessment, controls and evidence, not by a "certification" label.

17. Reference Sources

The following sources were used to verify the compliance guidance in this document. They are authoritative HHS/OCR materials; contractual and legal conclusions should still be confirmed for the exact Van-Kaal/ZocDoc relationship.

- **HHS — Summary of the HIPAA Security Rule** — <https://www.hhs.gov/hipaa/for-professionals/security/laws-regulations/index.html>
- **HHS — Guidance on Risk Analysis** — <https://www.hhs.gov/hipaa/for-professionals/security/guidance/guidance-risk-analysis/index.html>
- **HHS — Business Associates** — <https://www.hhs.gov/hipaa/for-professionals/privacy/guidance/business-associates/index.html>
- **HHS — HIPAA & Cloud Computing** — <https://www.hhs.gov/hipaa/for-professionals/special-topics/health-information-technology/cloud-computing/index.html>
- **HHS — Breach Notification Rule** — <https://www.hhs.gov/hipaa/for-professionals/breach-notification/index.html>
- **HHS — Minimum Necessary Requirement** — <https://www.hhs.gov/hipaa/for-professionals/privacy/guidance/minimum-necessary-requirement/index.html>
- **HHS — Appointment reminders / smaller providers FAQ** — <https://www.hhs.gov/hipaa/for-professionals/faq/smaller-providers-and-businesses/index.html>
- **HHS — Security Rule certification FAQ** — <https://www.hhs.gov/hipaa/for-professionals/faq/2003/are-we-required-to-certify-our-organizations-compliance-with-the-standards/index.html>

Source basis: ZocDoc No-Show Reduction Pilot — Process Specification & Technical Requirements, prepared by Gokul Chandran, last updated 16 September 2026. The current specification targets a 25–35% no-show reduction, a 90-day pilot, live reminders and IVR, PostgreSQL storage, encryption, practice access and a Sep 28 presentation. The proposed API-first model in this document is an evolution of that workflow.